

Parameters and architecture

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Forward passes by hand

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Activation functions

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Representing inputs as vectors

Without opening the old lesson, explain one key idea from Representing inputs as vectors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Sort the quantities for Parameters and architecture. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Parameters and architecture. Proposed explanation: “Count one bias per input”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Count a new architecture independently

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the output layer, with one bias per unit. Give each layer count and the total. If hidden width becomes 4, recount and explain why activation values are excluded.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Count a single layer

A dense layer has 4 inputs and 2 outputs with one bias per output. Count weights, biases and total.

My reasoning and evidence

6. W2 Sort the quantities

Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, parameter, architecture choice or training setting.

My reasoning and evidence

7. W3 Change the bias convention

Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.

My reasoning and evidence

8. W4 Count a new architecture independently

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the output layer, with one bias per unit. Give each layer count and the total. If hidden width becomes 4, recount and explain why activation values are excluded.

My reasoning and evidence

9. W5 Limit a size claim

A team says its 26-parameter model must outperform a 14-parameter model. Give a fair check and one reason the claim is unsupported.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled parameters and architecture evidence sheet to Image-pattern Network Investigation. Show how you can distinguish architectural choices from parameter values. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Parameters and architecture

Count trainable weights and biases separately from activation functions.

Dense toy architecture 2 inputs → 3 hidden units → 1 output. Every hidden and output unit has one bias; hidden activation is ReLU.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Trainable parameter count

	weights	biases	total
input to hidden	6	3	9
hidden to output	3	1	4

Read a cell using both its row and column labels.

Parameters and architecture

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count first layer

$2 \times 3 = 6$ weights plus 3 biases gives 9 parameters.



2. Count output layer

$3 \times 1 = 3$ weights plus 1 bias gives 4 parameters.



3. Total

Total trainable parameters $= 9 + 4 = 13$; ReLU adds no parameters here.

CHANGE ONE THING

Increase hidden width from 3 to 4.

Explain the changed result using the matching lesson.

How many new parameters did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Count trainable weights and biases separately from activation functions.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Total trainable parameters= $9+4=13$; ReLU adds no parameters here.

1. Total
2. Count output layer
3. Count first layer

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Total Total trainable parameters= $9+4=13$; ReLU adds no parameters here. Count one weight for each input-to-unit connection and one bias for each receiving unit. Two inputs connecting to three hidden units make six weights. The three hidden outputs connect to one output unit through three more weights. ReLU applies a fixed function and adds no adjustable parameter here. Architecture determines the connection pattern; learned values occupy those connections.

Practice 2: explain the deciding evidence

Changed-case practice: Increase hidden width from 3 to 4. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.

3. First layer 12 parameters; output layer 5; total 17.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: First layer 12 parameters; output layer 5; total 17. Each new hidden unit adds 2 input weights, 1 hidden bias and 1 outgoing weight.

Repair a missing building block

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y ? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original inputs cannot be distinguished by this network's final output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Representing inputs as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical inputs? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
